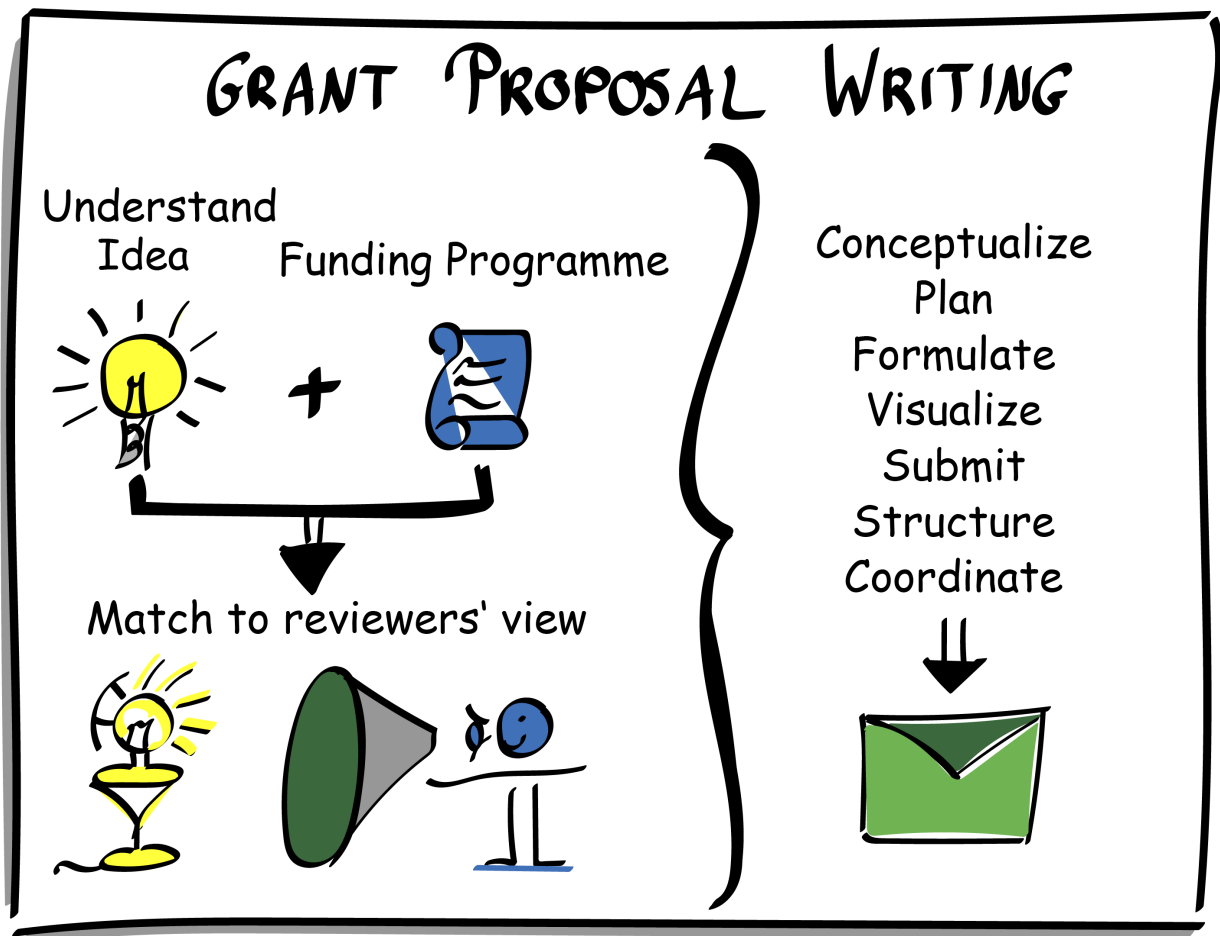


# Grant Proposal Writing

## How to design and communicate your project convincingly

### Handbook to the workshops



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## Legal notice

All examples in this training document have been generated by reference to open or already closed calls at the date of preparation. Conditions, formats and contents may change from call to call or may be updated by the funding authorities any time. Therefore, in preparing your own grant proposal you need to refer to the official documents available for your call of proposals.

## Imprint

Dr. Sabine Preusse (Text and Images)

RaumZeit e.K. Coaching Beratung Training

Mühlweg 1

61250 Usingen

Germany

Contact:

Email: [sabine.preusse@raum-zeit.de](mailto:sabine.preusse@raum-zeit.de)

Phone: +49-6081-9857005

Visualisations Dr. Sabine Preusse

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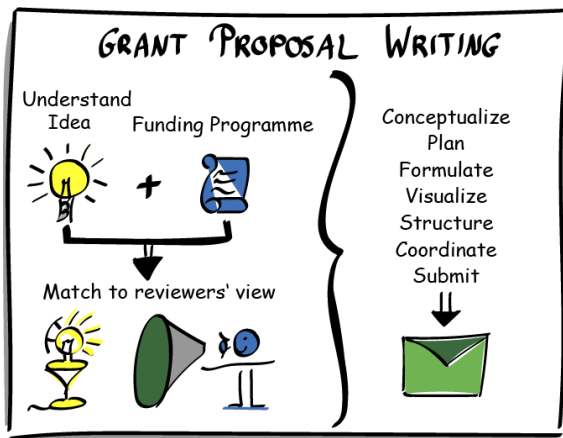
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## 1 General remarks about grant proposal writing



**Figure 1 Grant Proposal Writing**

Successful grant proposal writing is a fundamental basis to fund your career in research or to fund research and innovation projects outside of academia. There are plenty of national and international funding programmes with different funding schemes. Each scheme has its individual context and thematic focus, eligibility and funding rules, submission, and evaluation process. Thus, the detailed features and aspects of a successful proposal in one funding scheme naturally differ from those of a successful proposal in another funding scheme. However, the process of how to work out these details is a common denominator to the

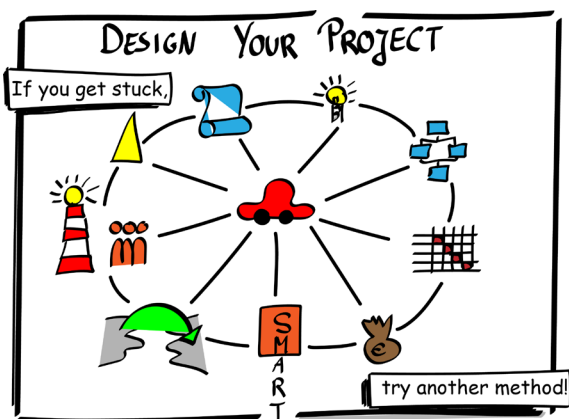
different funding schemes. It requires skills way beyond the ability to work out a research idea in detail.

Figure 1 gives a general overview of the overall process involved. You will start either with an idea for a project or a funding programme which is very interesting. There are many different methods, which help you to develop your idea into a work plan and develop all the details which you will need for writing up the proposal. For an excellent grant proposal, it is necessary to match the research idea with the reviewers' view on what is wanted by the funding scheme and what is considered fit for funding.

This handbook will give you an overview on research projects in general and then introduce different methods and aspects, which are helpful for developing a detailed project plan and to write it up in a grant proposal. However, it does not provide background information on funding programmes, cultural differences in grant proposal writing and differences with respect to the different disciplines. Please make sure, that when you are preparing your own proposal, to find out about these by talking to experienced grant proposal writers, consultants or even reviewers.

Finally, before we start you need to know the most important rule of proposal writing:

Always refer to the actual documents and guidelines provided for your call of interest.



**Figure 2: The RaumZeit toolkit for grant proposal development**

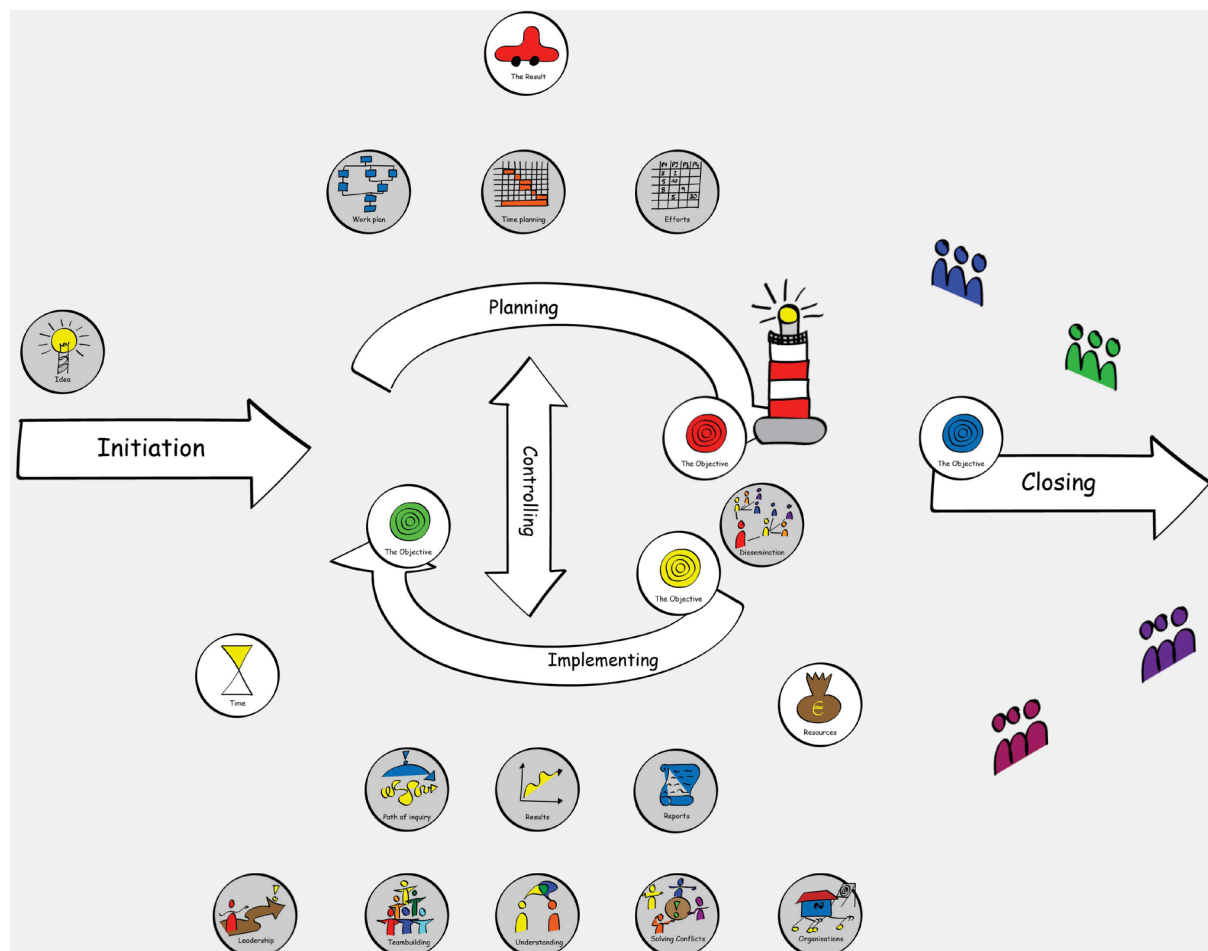
Whatever is written there prevails all information in this or other non-official documents.

### 1.1 How to work with the RaumZeit toolkit

The methods introduced in the workshops and in this handbook help you to look at the overall objective of your research project from different perspectives and to work out all the details that you will need to justify your budget and to write a convincing proposal (Figure 2). For an efficient proposal development process, it is helpful to change the method when you have the

feeling that you got stuck. It is also helpful to work with one mindset at a time. For example, if you are looking for details to make your objectives specific and measurable, it is helpful to just assume, that they are extremely relevant. What exactly makes them relevant will be the target of another method.

## 1.2 What is a research project?



**Figure 3 Research projects - an overview**

A research project targets at generating new knowledge, methods, tools, devices, algorithms, theories, evidence and much more within a specific time and with limited resources. Following from the idea to the end, a project evolves in four phases (following the definition of Project management Institute, Inc., 2013)<sup>1</sup>:

- During the **initiation phase** you develop your idea into an initial concept and identify a funding programme in which this project will have a significant chance of success. At the end of this phase, you will decide to submit a proposal.

<sup>1</sup> **Project management Institute, Inc. 2013.** *A guide to the project management body of knowledge (PMBOK Guide)*. 5th Edition. Pennsylvania : Project Management Institute, Inc., 2013.

- During the **planning phase** you anticipate the implementation of the project and plan all activities and resources in detail. At the end of this phase your project has been submitted. If it is accepted for funding, then this phase will end with the signature of the contract and the official start of the project.
- The contract defines the first and the last day of the **execution phase** during which you implement your project. In general, this is also the time for which you can charge the costs to the funding agency.
- Finally **closing** the project incorporates the final reporting and finishing of all activities.

All these phases evolve around the **overall objective**, which is defined by an **overall result**, which will be available at a given point of **time** and the **necessary resources** (Figure 3).

### Planning

The planning of the projects starts with a rough idea about the results and leads to SMART (specific, measurable, attractive, realisable, terminated) objectives which are fixed with the grant proposal and the contract with the funding body. The details on how to achieve the objective are provided in the work plan and in the budget plan.

At first sight, your detailed plan contains information on how you plan to do the research, trouble shooting of technical or scientific problems, results of each individual step in the work programme, reporting and publishing and dissemination of the results. However, implementing a research project successfully requires more: Someone who is willing and able to lead, teambuilding so that collaborators and staff can work together efficiently, a common understanding of about everything especially in interdisciplinary or transdisciplinary research and in collaborative projects with the industry, conflict resolution, and organisations which allocate staff and resources to the project. All these activities require resources which need to be planned as a part of the scientific and administrative project management. The latter has either a dedicated work package in the work plan or is part of the scientific workload.

### Implementing

However, once the project has started it is run by persons who are part of organisations. As time goes by interests and understanding of what is exactly needed change and thus the result will be adjusted within the frame of the contractual SMART objective. Thus, executing the project has a lot to do with leading and motivating collaborators, integrating the institutions' strategic objectives, problem solving, measuring progress and success, negotiating roles, tasks, approaches and benefits and supporting the project team to work together efficiently. Monitoring and **controlling** allows for a good progress reporting and justification of these changes to the plan.

### Closing

Once the project is officially terminated, the final reports are written and hopefully accepted. This is the administrative essence of the closing phase.



**This is about work but what makes the project attractive?**

All this is only about what is happening between the first and the last day of the research project. It is all about the overall objective and thus about the overall result. However, it does not make the overall result attractive. For being attractive the project needs to have an impact. This means, one or more target groups pick up the result in whatever they are doing. As they can now do something they could not do before, they will have a benefit. And this benefit accumulated over a given time is the impact.

Impact does not arise by its own. Dissemination and exploitation activities act like a beacon to make target groups interested in exploiting the research results. These target groups can be very different: The project partners, the individuals on the projects, the scientific community, society, industry, politics, the interested public... It depends on your research project and it depends on the funding body issuing the call for proposals. For funding bodies, the investment in research projects is a strategic instrument to implement their mission. These missions differ and thus it differs, what they consider an attractive impact.

## 2 Identification of the right funding scheme

There are plenty of national and international funding programmes. Sometimes you find a programme which is perfect for your research project. However, it is more likely that you need to adapt your project idea or an already existing research project concept so that it fits nicely. As a third variation you will have the funding programme before you are developing the idea. You can be successful with any of these scenarios if you understand the call for proposals and then match your project to it so that it fits nicely.

You can find a call for proposals by browsing the world wide web or targeted funding institutions on a regular basis, subscribing to email newsletters, visiting information events, asking others who have an overview. Once you have found an interesting funding programme you want to get a general understanding of it first and then decide whether you are going to look for the details. The following general information will help you to orient yourself.

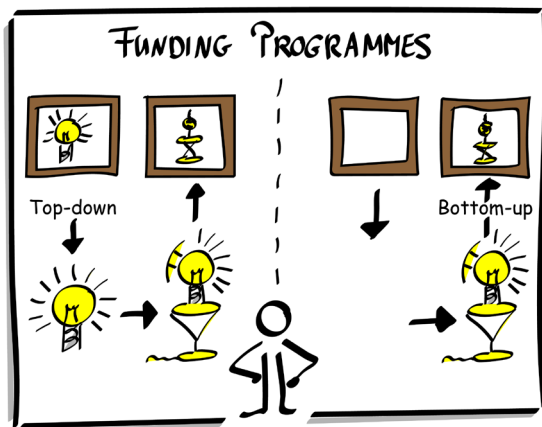
### 2.1 Funding Programmes

Public funding of research and innovation projects is a strategic instrument to bring about strategic objectives of the funding body or institution. These objectives can be of general nature (fund nice and feasible research projects) and they can be very explicit (implement a political agenda). You will find background information on the webpages of the institution, in vision and mission statements, in the official documents, in strategy papers, in political agendas and so on. Public funding is issued by national ministries, dedicated funding bodies, foundations, the European Union and governmental or non-governmental organisations.

Besides differences in the objectives of the investment into research and innovation, funding programmes differ in many aspects:

- Objectives
- Themes and topics
- Eligibility criteria (who may apply or receive funding)
- Funding rules
- Duration of funding
- Funding limits
- Evaluation process
- Contractual rules

However, all funding programmes can be sorted according to how the topics of research projects are defined:



**Figure 4 Funding programmes are top-down or bottom-up**

application area and to ask researchers to propose interesting and attractive research projects. As the applicant takes over the expert role of defining what exactly is relevant in his or her field of research, this process is called bottom-up (Figure 4 right side).

For designing a top-down programme, funding agencies involve external experts. In this process lobbyists have taken an influence on the objectives of the different calls of these programmes. For you as the person writing the proposal, this has an advantage: The reviewers have been instructed before, which aspects the funding authority considers attractive for funding and you do not have to justify this anymore. However, sometimes individual calls seem to expect a specific proposal by a specific consortium. Therefore, the challenge for a successful proposal lies in deftly answering to the expectations which are defined in the respective call for proposals.

In contrast, bottom-up programmes allow you to freely define your research objectives, the solution and impact. The challenge here lies in persuading the reviewers of the importance and attractiveness of the proposed research topic. In this case knowledge about the mission of the funding agency helps a lot.

The following questions help you to identify what kind of programme you are looking for when screening for suitable funding opportunities:

- Can I do my research project all by myself? Look out for individual fellowships.
- Is it basic research? Look out for funding programmes targeting basic research.
- Is it applied research? Look out for funding programmes requiring the participation of enterprises.
- Do I need only national organisations to implement the project? Look out on national level.
- Do I need to collaborate internationally? Look out for funding targeting international collaboration.
- Is there a top-down call, to which I can easily fit my project?
- If there is no top-down call, which bottom-up call is suitable?

## 2.2 Information, contact points and programme execution organisations

There are different types of organisations providing professional information on funding programmes:

- There are institutions surveying funding opportunities, extracting the most important information and publishing these to their clients in information events, on the website or in newsletters. These institutions are acting as multipliers for the various funding programmes.
- For European funding programmes, National Contact Points exist in all member states and associated states. They provide up to date information and support before proposal submission and during the implementation of the project. In Germany they are organised around specific funding themes or funding programmes.
- Programme execution organisations (in German Projektträger) are charged by a ministry to run the overall process of publishing the calls, organising the evaluation process, evaluation of proposals and decision-making.

For being alert what happens on the funding market it makes sense to subscribe to the newsletters of those institutions, which cover the funding programmes of interest. Once you have found a suitable funding programme, acquaint yourself with the procedures, processes and deadlines and refresh this knowledge on a regular basis. Then it will be easy to keep an overview and get started.

## 2.3 The funding scheme fingerprint

Once you have identified an interesting call for proposals you should gather all official documents concerning exactly this call. Look out for something like the following documents:

- A document announcing the call and providing the basic data such as deadlines, scope or topic, eligibility criteria etc. It is often called Bekanntmachung, Aufruf, Call Fiche, Call for Proposals. Sometimes this information is presented just on a website.
- Guidelines providing in depth explanations on the proposal submission and evaluation process, i.e. Leitfaden, Anmerkungen, Guideline.
- Templates for the proposal, official documents, budget calculation etc.

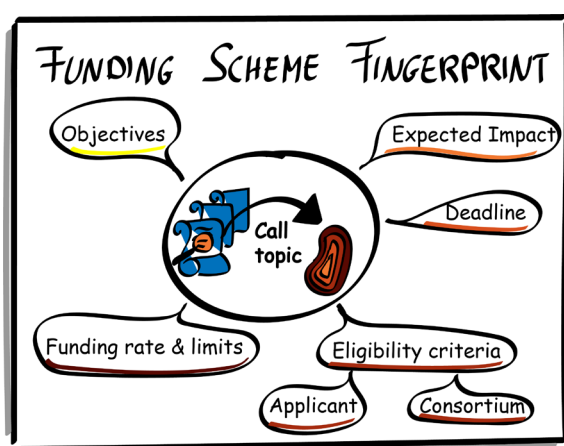


Figure 5 Funding scheme fingerprint

If you read all these documents, you will be lost in detail very easily. The *funding scheme fingerprint* (Figure 5) will help you to focus on the relevant information for a qualified decision if it makes sense to prepare the proposal or not. Furthermore, it defines what a successful proposal will look like and which information it needs to contain.

The funding scheme fingerprint is hidden in the official documents and you obtain it by meticulously reading each document and writing down the important words and passages for later use. The information you should gather in this process is summarised in Figure 5:

- The call topic is the title of the call for proposals or just the name of the funding programme. In European funding, each call for proposals also has an identifier, which will allow you to find the right entry point to the submission portal. It is helpful to collect both.
- All funding programmes formulate objectives for the research projects they are looking for. In a top-down programme, they are explicitly defined in the call topic and in the corresponding work programme, which defines the contents of the funding programme. In a bottom-up programme, they are often very general but likewise important.
- In top-down programmes the impact may be clearly defined. In bottom up programmes it is more general. Impact may target different levels and you should collect all of them.
- Finally, administrative criteria such as the deadline, the detailed eligibility criteria concerning the applicant and the consortium, the funding rate and funding limits are part of my funding scheme fingerprints.

For extracting the fingerprint, just browse the documents for the most important information and discard all the details.

You can now check thoroughly if your project idea, the applicant/the principal investigator (PI) and the consortium are eligible for funding in this programme under this call. If this is unclear, you should contact an official authority to clarify the points in questions. Otherwise, you will run into danger to be kicked out of the evaluation process even before the project itself has been assessed.

Once you have decided that this funding programme is really an option, you should go back to all the documents and add details to the fingerprint to turn this into what I call the *frame*. Copy the keywords and passages word by word to a mind map. This will be very helpful in writing up your proposal as you can use formulations the reviewer already knows and explain how you are going to bring it about. Furthermore, you can use it to check whether your proposal contains all the required information once you have written up everything.

## 2.4 Mind mapping

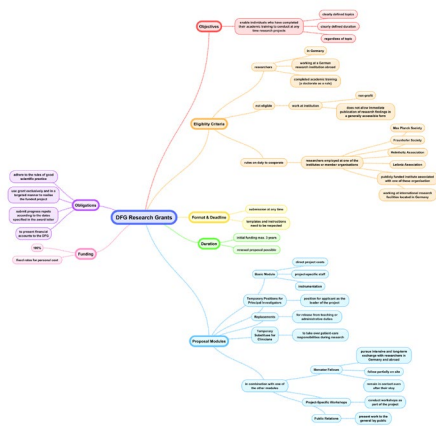
You can use a mind map to creatively note ideas. As a diagram visually outlining information it was already used in the 3<sup>rd</sup> century by Porphyry of Tyros. It was made popular by Tony Buzan in his BBC TV series *Use Your Head* in 1974 (Wikipedia<sup>2</sup>). Figure 6 shows a funding scheme fingerprint done in a mind map.

The overall process is simple:

- Write your topic into the centre and add your ideas radially around it (branches). If you have ideas with respect to these ideas you add them to the respective node with sub-branches.
- Use different colours.
- Link and cross-link ideas in order to show relationships

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<sup>2</sup> Wikipedia contributors. (2020, May 7). Mind map. In *Wikipedia, The Free Encyclopedia*. Retrieved 10:12, May 21, 2020, from [https://en.wikipedia.org/w/index.php?title=Mind\\_map&oldid=955425210](https://en.wikipedia.org/w/index.php?title=Mind_map&oldid=955425210),



**Figure 6 Mind map of funding scheme fingerprint**

state-of-the-art for your next paper or your thesis from this. When preparing a proposal, you might wish to do the same.

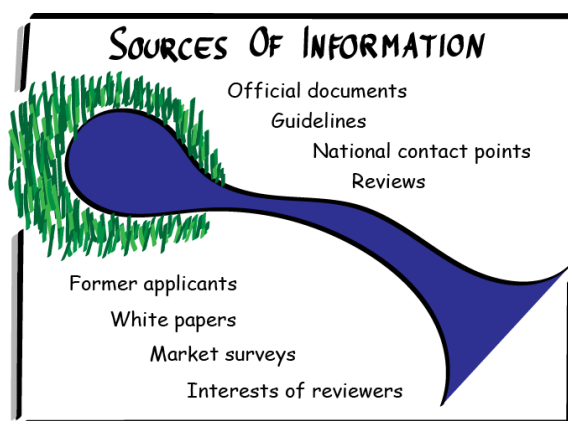
- Organising yourself with a fancy and structured to do list, which can be easily detailed in those parts where needed.

## 2.5 Sources of detailed information

At the beginning and during the proposal writing process, different sources of information play an important role for understanding what exactly is wanted and expected by the funding authority (see Figure 7). They can be classified as direct sources and indirect sources.

Direct sources are those published or provided by the funding authority and its representatives. To these belong the official documents of the call for proposals, official guidelines, and expert knowledge from the official contact points, e.g. the National Contact Points or “Projekträger”. If your proposal is rejected, evaluation summary reports will provide further valuable information. This can be weak points or hints to what the reviewer did not understand.

Former applicants, white papers, market surveys and interests of the reviewers can be considered as indirect sources of information. This information often relates to the objectives of the funding



**Figure 7 Sources of information**

programme or the mission of the funding authority. Then you better understand what is meant with the way objectives have been formulated, the context and the political strategy behind. Sometimes indirect information “just” provides insights about what is important to reviewers, challenges in the decision making, and things happening “backstage” during an evaluation process.

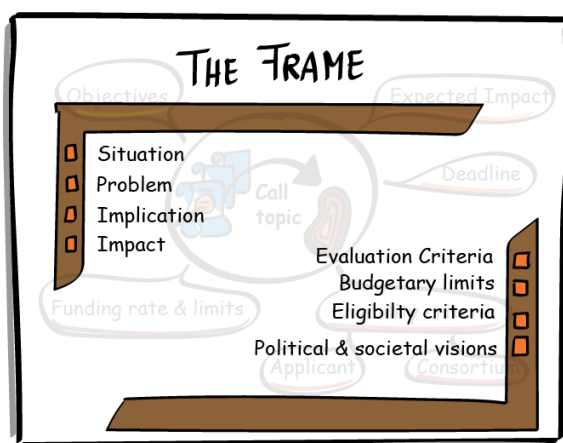
The following list just indicates what and where you can look for information:

- Databases of running and completed projects

provide valuable information for the positioning of the project idea with respect to state-of-the-art and the funding mission of the funding authority. You will find the links on the websites of the funding authorities.

- Publications and studies published by the funding authorities provide information on their strategic vision.
- Market surveys can be found in the WWW. Some of them are free of charge.
- Strategic Research Agendas of the European Technology Platforms or Joint Technology Initiatives.
- National and international Patent Databases. One outstanding example is Espacenet of the European Patent Office.
- White Papers of governmental authorities.

### 3 Assessment of the framework defined by the funding scheme



**Figure 8 The Frame**

There are must have and nice to have aspects which your project should fulfil to increase your chances to success. Some of them are already contained in your funding scheme fingerprint, most not. Therefore, you add now all the relevant details from the official documents and guidelines while scrutinizing them.

The result is a very detailed mind map which I call the *frame* (Figure 8). If you design your project exactly to this frame it will fit nicely to the requirements of your call for proposals.

The frame is provided by the funding programme and consist of two important parts:

- Part 1 defines the situation to be considered, the problem, its implications and the impact this project is expected to achieve. This part defines what is important from the scientific and technical point of view.
- Part 2 contains evaluation criteria, budgetary limits, eligibility criteria and political & societal visions. This part provides the criteria, which will turn an excellent scientific approach into a winning proposal.

A successful proposal presents a project vision, which fits the complete frame nicely. It picks up the defined situation, the problems which should be solved and offers a solution which will take care of the implications and which will at least result in the requested impact. You can use Part 2 of the frame to focus the setting of the scientific approach and to make the project attractive for the funding authority.

In my experience Part 1 is often easy to handle. Part 2 seems to be somewhat diffuse and administrative; therefore, it is often neglected especially as it is difficult to relate the project to political and societal aspects. However, answering clearly to Part 2 of the frame will help the reviewer to rank two proposals with an equally attractive scientific approach.

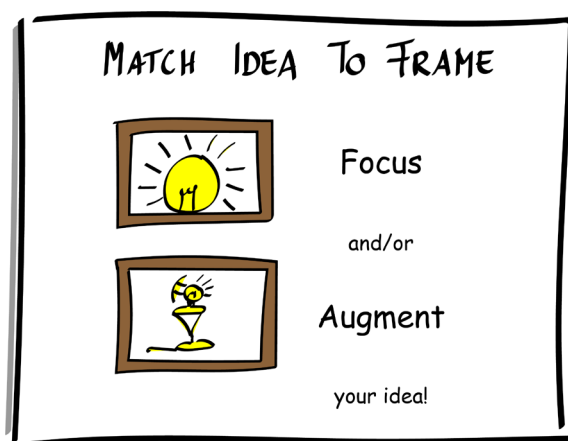
#### 3.1 Mapping the research idea to the framework to achieve a high relevance

You can use the frame to inspire a brainstorming for

- developing the project objective,
- turning your idea into a first work plan,
- developing lines of argument for the impact or justifications for why you and your consortium should be funded.

If you use an electronic mind map you should discriminate your ideas and comments from the keywords copied from the official documents.



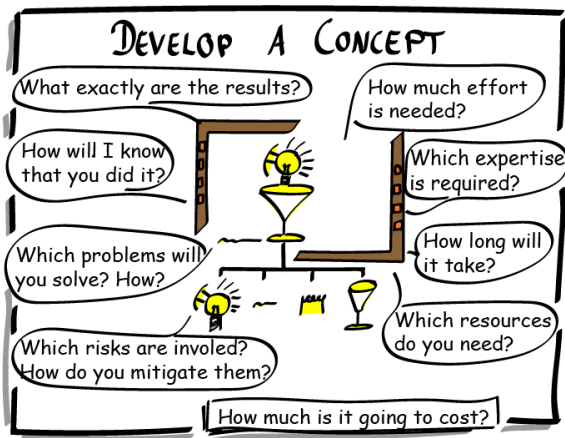


**Figure 9 Match idea to frame**

If you do a brainstorming on objectives and work plan without a funding programme in mind or if you adapt your project to a new funding scheme, the frame will now provide the background to design your project to the funding programme (Figure 9):

- Is anything missing, that is required by the funding programme? Have you included enough and appropriate dissemination, exploitation, training, standardisation, or project management activities? They are an important part of the impact or of the evaluation criteria.
- Which research aspects are too much? Focus your project so that it fits nicely to the scope, duration, and budget, by carving it out of your original research design.
- Is your project too small? Augment it by adding relevant research questions or by embedding it in a bigger scientific or more applied context.

## 4 Developing a concept and/or research design



**Figure 10 Ingredients of a concept**

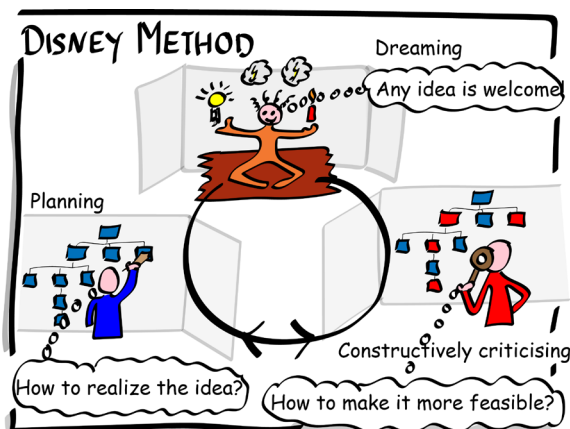
A grant proposal is much more than the outline of a research idea: It provides a recipe of what needs to be done in order to achieve the overall objective in the given time and with the planned budget. Therefore, a good grant proposal includes a detailed concept or research design, which answers the following questions (Figure 10):

1. What exactly are the results?
2. How does someone, who was not involved in the project know, that you have been successful?
3. Which problems will you solve? How?
4. Which risks are involved? How do you mitigate them?
5. How much effort is needed to do all the work?
6. Which expertise is required?
7. How long does it take?
8. Which resources do you need?

The first five questions lead to the definition of the results of the project and the work plan. Questions five and six target at identifying the right stakeholders, team members and project partners. Questions five, seven and eight are the basis for the budget calculation.

### 4.1 The Disney method

For developing research designs of any size, projects of any size, business plans of any kind or personal career plans, the Disney method (Figure 11) proved to be very powerful in structuring your design process.

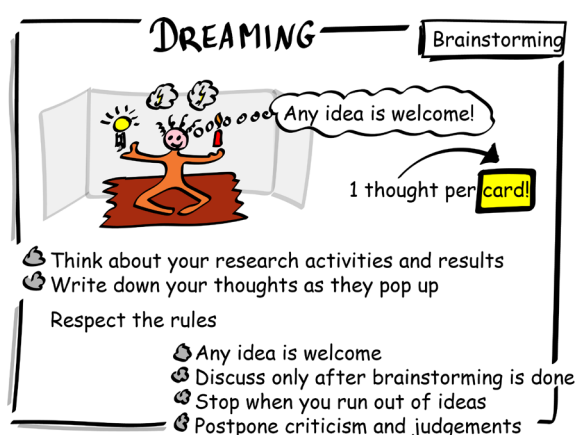


**Figure 11 Disney method**

This is part of the Wikipedia description of it:

*“The Disney Method, developed by Robert Dilts in 1994, is a complex creativity strategy in which a group uses four specific thinking styles in turn. It involves parallel thinking to analyse a problem, generate ideas, evaluate ideas, construct and critique a plan of action. The four thinking styles are - outsiders, dreamers, realisers and critics.*

*In the first thinking style the group think as outsiders to gain an analytical, external view of the challenge. They then act as dreamers to brainstorm ideal solutions.*



**Figure 12 Disney method - dreaming**

*They use divergent thinking to conceive creative and radical ideas.*

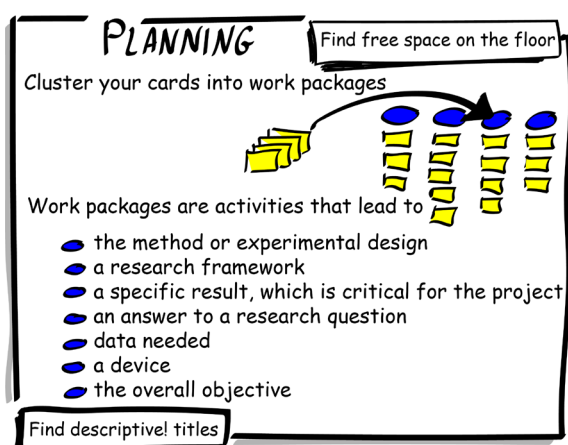
*In the next mode the group adopts a realiser viewpoint. They act as pragmatic realists and use convergent thinking to review the ideas left by the dreamers. They select the best idea and construct a plan for it.*

*The fourth viewpoint is that of critics. They review the plan made by the realists in order to identify weaknesses, obstacles or risks. They seek to improve the plan.*

*The method is comparable to and an alternative to de Bono's method of the Six Thinking Hats." (Wikipedia<sup>3</sup>)*

Sounds complex but it is easily implemented into your everyday life. In case of grant proposal writing it works as follows:

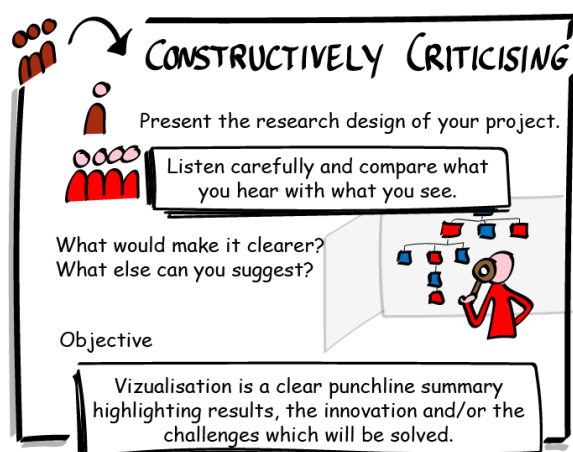
1. **Dreaming** (Figure 12): Brainstorm your overall research questions or problem and write down all ideas on cards. Remember: Any idea is welcome even if it is completely unrelated to your project! We are looking for quantity, not assessing the quality as criticism and judgement are postponed to criticising. When you are running out of ideas, move on to planning.
2. **Planning** (Figure 13): Imagine yourself as the realiser. Your task is to find a way to implement the project. Criticism and judgement are postponed to criticising and not your business right now.



**Figure 13 Disney method - planning**

However, you are welcome to add ideas, which you will need to efficiently implement the project. To do so efficiently, take the cards from the dreaming, find an empty space on the floor, and cluster them. When writing a grant proposal this is the concept development phase. Results of this phase are clustered cards, a first work plan or Pert diagram, a Gantt-chart and a first effort planning. In everyday time management the result of this phase is the structuring and prioritising and timing of your tasks. If you need more ideas move back to dreaming. If you have a plan move forward to

<sup>3</sup> Wikipedia contributors. (2017, September 11). Disney method. In *Wikipedia, The Free Encyclopedia*. Retrieved 19:21, April 1, 2020, from [https://en.wikipedia.org/w/index.php?title=Disney\\_method&oldid=800072172](https://en.wikipedia.org/w/index.php?title=Disney_method&oldid=800072172), license cc-by-sa 3.0 unported, [https://en.wikipedia.org/wiki/Wikipedia:Text\\_of\\_Creative\\_Commons\\_Attribution-ShareAlike\\_3.0\\_Unported\\_License](https://en.wikipedia.org/wiki/Wikipedia:Text_of_Creative_Commons_Attribution-ShareAlike_3.0_Unported_License)



**Figure 14 Disney method - constructively criticising**

criticising.

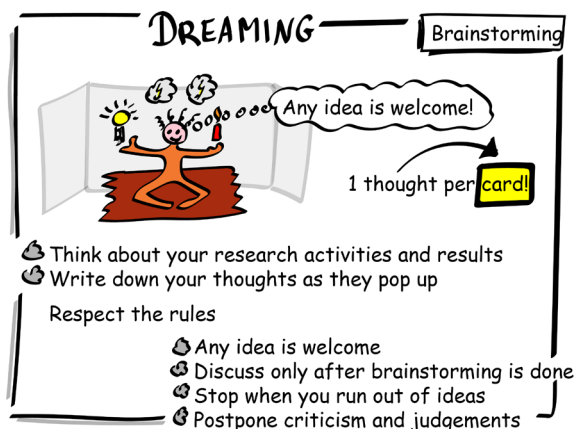
3. **Criticising** (Figure 14): Now activate your inner critic. It is your task to highlight any flaw in your plan and to suggest how to improve it or even turn it into a strong point. If you need some ideas with respect to the suggestions move on to dreaming and brainstorm about this problem. If you can directly incorporate the suggestions into the plan move on to planning.

If your plan is good enough you can write it up in your grant proposal. If the proposal is of high quality, you can submit it to the funding agency.

The Disney method will become very efficient if you

- use different places which support you in stimulating creativity, planning, and criticising.
- allow your cards to rest on the floor or on a board until you have done all the detailing, clustering, and structuring.
- put all the details you have in mind on the cards, one detail per card. Though this seems to be too many cards, you will be able to move them around flexibly. If everything feels good, you will only have to formulate sentences around the words on the cards. This will quickly start you writing.

## 5 Dreaming with brainstorming



When you need ideas for any topic, brainstorming is the simplest method to move along (Figure 15):

- Define a topic.
- Write down all ideas as they pop up.

For writing down your ideas, you can use workshop cards, post-its, a paper, a mind-map, a note pad or whatever is available for taking notes. However, to make it work well you should respect the following rules:

**Figure 15 Disney method - dreaming**

• Any idea is welcome: If you brainstorm on the contents of your next paper and the word coffee pops up you will write it down along with all the other ideas. This is very important for stimulating the creativity of your brain and to allow for new and unconventional ideas.

- Postpone all discussions: Evaluating and questioning the ideas when they come up is a certain killer of your creativity. You will do so after the brainstorming.
- Stop when you run out of ideas or when you realise that you repeat yourself.
- It is brainstorming now and criticising or judging later. Only if you postpone criticism and judgement you will get into the flow of generating new ideas. By not respecting this rule, you might discard good ideas because they seem to be crazy and not feasible without thinking them through.

When you brainstorm you should free yourself of having to structure the ideas right away. Therefore, try to work with cards or with a mind map:

- **Cards:** Write down one idea per card and put them unsorted in front of you so you can look at them while brainstorming. When you are done you take the cards and cluster them. A pin board is nice, but you will have to use needles. I use the empty space on a floor or a table to flexibly move around my cards until I will have found the right structure.
- **Electronic mind map:** By adding ideas and refining them you are already doing a semi-structured brainstorming. You can change the structure easily until it seems to be the right one.

Brainstorming works for a single person or in groups of various sizes. For groups, you can use the following variants:

- **Individual brainstorming:** Each person privately notes the ideas on workshop cards. When all run out of ideas or time is over, each person presents his or her cards and pins them to a board. You add ideas that come up during the presentation. Then you cluster the cards together. If it is important, that ideas will be contributed anonymously, the moderator collects the cards and shuffles them before pinning them to the board.

- **Brainstorming in the group:** Brainstorm simultaneously by calling out the ideas which are then written down by the moderator. As others can associate with the ideas on the board, new ideas will come up easily.

Brainstorming is very efficient and suitable for problems of any kind, be it technical, scientific or personal.

Figure 16 and Figure 17 show some questions for dreaming your research project:

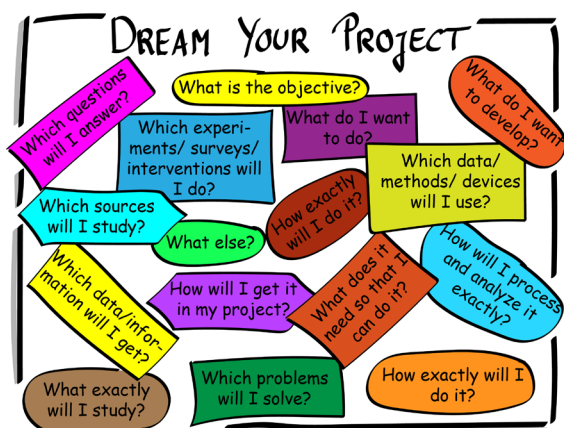


Figure 16 Questions for brainstorming 1

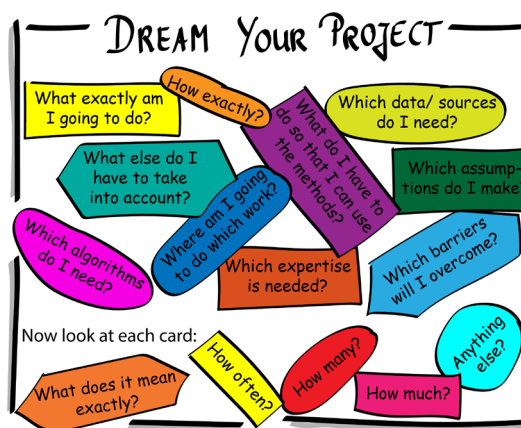


Figure 17 Questions for brainstorming 2

## 6 Planning

Planning a research project is a complex undertaking. The first draft of your work plan is the result of the brainstorming. It needs to be refined iteratively until you have enough details for writing a good proposal.

### 6.1 Elements of a work plan - objectives, results, milestones, work packages, tasks

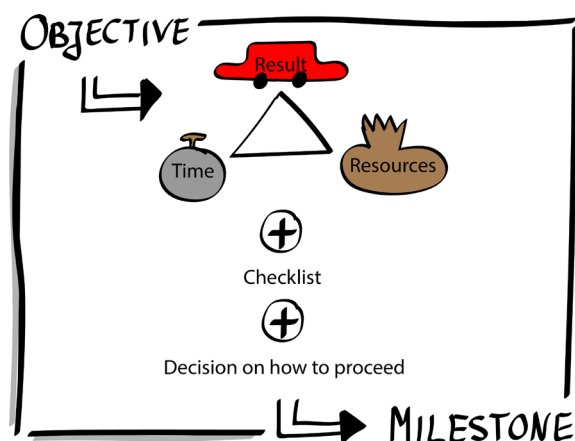


Figure 18 Definition objective

Objectives are the key to any project and research collaboration.

An **objective** defines which specific **result** will be achieved until a given **time** by using certain **resources**, e.g. an estimated amount of work time (effort!), consumables, equipment, travel and other costs.

Results in research projects can be knowledge, data, models, simulation results, code, lists with parameters, algorithms, tools, methods, materials, devices etc. They are documented in different ways such as reports, publications, prototypes, manuals, guidelines, books,

and monographs.

It is very important to define a result in such a way that a third person can approve its existence. This is achieved by specifying the result in great detail: which knowledge is expressed, which performance it will show, which features it has, which questions it will answer, on which data or sources it will be based and by which documents it will be supported.

A **milestone** is a point of decision-making in the project. It consists of an objective, a checklist against which the result is being reviewed and a decision on how to proceed.

The decision-making is formally done in a milestone review meeting.

**Work packages** are defined to break down the work in the project into smaller manageable units leading to an intermediate objective.

Work packages present the major activities in a work programme, such as development of the research framework, development of the required method, running of a specific type of experiments, doing field work in a specific place, considering a specific time period in history, analysing data and sources with respect to a major research questions or hypothesis and so on.

**Tasks** are the major smaller activities structuring the work in a work package.



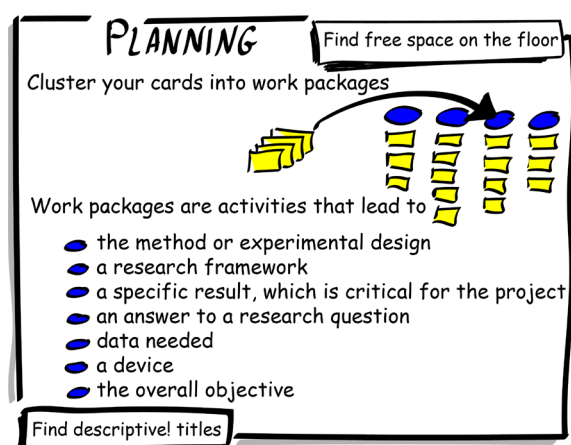
## 6.2 Visualising the work plan – the “PERT” diagram

At the end of the dreaming phase you have a pile of cards with ideas. The objective of the first round of planning is to develop a first version of your work plan and to visualise it as a research design. This shows the logical structure of your work plan. It is a less detailed version of a PERT (Program evaluation and review technique) diagram. Besides visualising the logical order of the development process with dependencies between the individual work packages, this diagram contains information on the project months in which the work packages start and end, their duration, and the time buffer available.

The visualisation of your work plan is a very powerful tool: You can use it to match your project to the funding programme of choice, discuss it with peers or potential partners, and to obtain a good structure for writing up the proposal. This is, how to get there after the dreaming phase.

### Step1: Cluster your cards

Before you follow the instruction, be clear about the following: Work packages are containers for work



that will generate an important result for the project. Therefore, many of our work packages will produce the results phrased in our objectives. All the other work packages are needed for a targeted research agenda or additional activities, which we want to put in the focus of communication. These could be dissemination and exploitation activities, project management, quality management, standardization or whatever makes sense in the context of your project.

In terms of size work packages are big enough to structure the project nicely and small enough to allow for controlling the progress of work.

**Figure 19 Disney method - planning**

Now, take your pile of cards and find a free space on the floor. Now cluster the cards into work packages (Figure 19). Usually you can group the different bits and pieces of knowledge around a distinct activity that will lead to a new method or an experimental design, a research framework, a specific result, an answer to a research questions, data needed, a device, the overall objective. These results are the output of activities such as literature research, modelling, experimental set-up, up scaling, demonstrating, validating, analysing and so on, which are the typical step stones of research projects on the way to the overall result. Therefore, they are the logical units of your work plan, the **work packages** (WP). In Figure 19 they are represented by blue and oval cards. The yellow and rectangular cards represent major activities which you need to do within the work packages. We call them **tasks** (T).

If you have clustered your cards in a first approach in research questions, methods, partners, experiments and other, you will need to shuffle them around into new clusters in order to get the structure of the work plan. A good guess is to start by using your research questions as work packages and then add all the development activities that you need to find the answers to the questions.



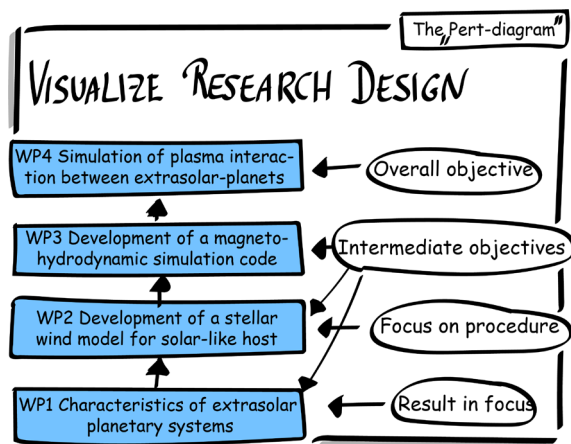


Figure 20 Visualizing the research design

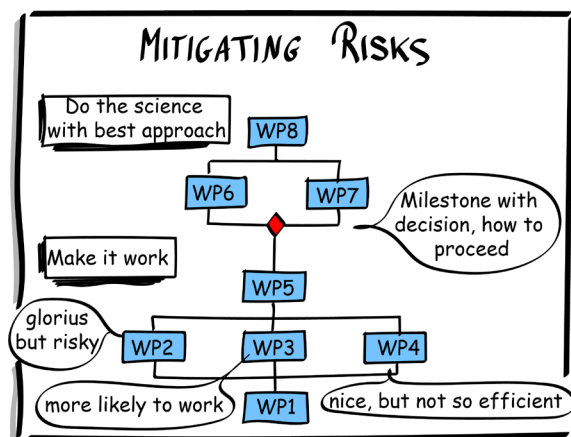


Figure 21 Research design - mitigating risks

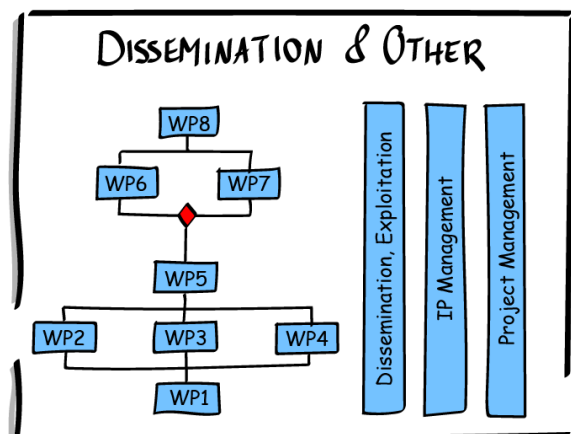


Figure 22 Research design – other activities

Often you will find that you have already been thinking in work packages and that you have hardly any tasks. In this case just do another brainstorming on the details of your work packages.

### Step 2: Find descriptive titles for the work packages

With some experience you will realize that research projects in your field of research usually follow a common logic. However, I use the logical structure of the work packages to develop a punch line summary of the project, of which I pick up parts as a line of argument in most chapters of the proposal. Therefore, I now formulate a descriptive title for each work package. The titles are specific on the output or result of the work package, or the challenge, which will be solved, or the innovation, which is achieved within. These descriptive titles are about a half sentence in length. If your first version is longer, it will be easy to make them more concise later.

### Step 3: Visualise your research design

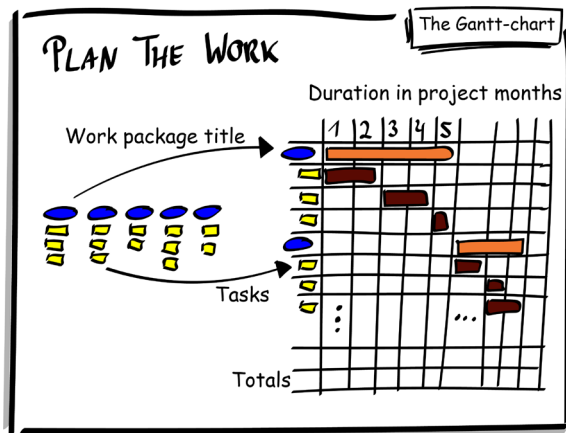
Now you sketch your work packages as boxes and indicate with arrows, how the generation of knowledge is processed during the implementation of your project. Boxes in a sequence (Figure 20) mean, that the result of the one work package is used as a starting point in the following work package. If you have activities which are independent of each other and which could be done in parallel, you can arrange your boxes in parallel. You can see this in Figure 21. It also shows a more complex research design, and how you can mitigate risks.

Some work packages accompany the whole project and thus run from the beginning of the project to its end. This is shown in the Figure 22.

### Some important remarks

Some reviewers will start evaluating your proposal by looking at the PERT diagram. If you are familiar with this kind of diagrams it can tell you if all important elements of a research project in a specific field of research have been incorporated. If you make sure that your work package names summarize what you want to do in the work packages you will provide all the information the reviewer is looking for. However, if it is not appropriate to visualize your research design, do not put the visualization.

### 6.3 Time planning with a Gantt chart



**Figure 23 The timeline**

The research design presents the logical order of your work. It does not contain any information on the duration of the work packages and the time dependencies. This detailed planning is most easily done and visualised with a Gantt chart (Figure 23).

You can use a table calculation programme. Write your work packages (identifier and full name) from your visualized research design in the lines of the first column. Then you add the tasks belonging to each work package in their logical order in the lines below each work package. In the columns following this information you add the project months. Now you

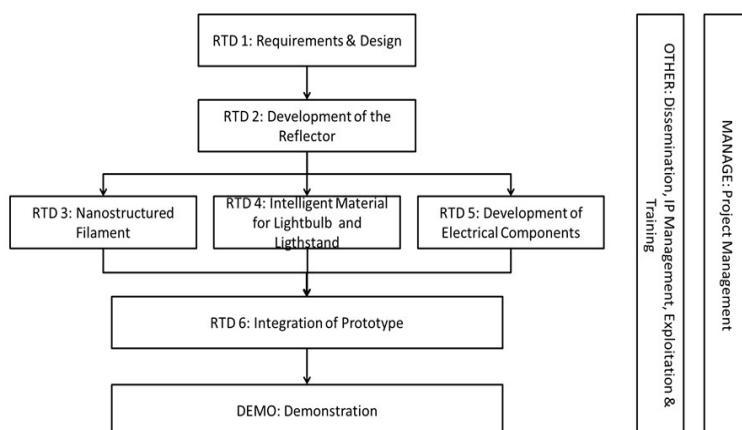
indicate the duration of each task by shading the respective project months. Finally, you highlight the total duration of each work package.

Figure 24 and Figure 25 show the research design for the Aladdin's lamp project and a Gantt chart. This also has milestones and planned reports. Furthermore, you can indicate which result of a task is used by which following tasks by adding arrows.

First proposals are often too ambitious in time and budget planning. If you are planning your own position you can prevent this by using the Gantt chart to get an idea if the workload is realistic or not. For this you estimate the efforts for each task in each month and write them in the shaded cells. By summing up all efforts for a month you can see if it is too much or not.

#### An important remark

The Gantt-chart as a timeline facilitates the reviewing of your proposal if you show the full names of work packages and tasks. If you only show the identifies, then the reviewer will need to flip the packages and refer to the written text in the work programme. This is an unnecessary effort.



**Figure 24 Pert diagram for the Aladdin's lamp project**

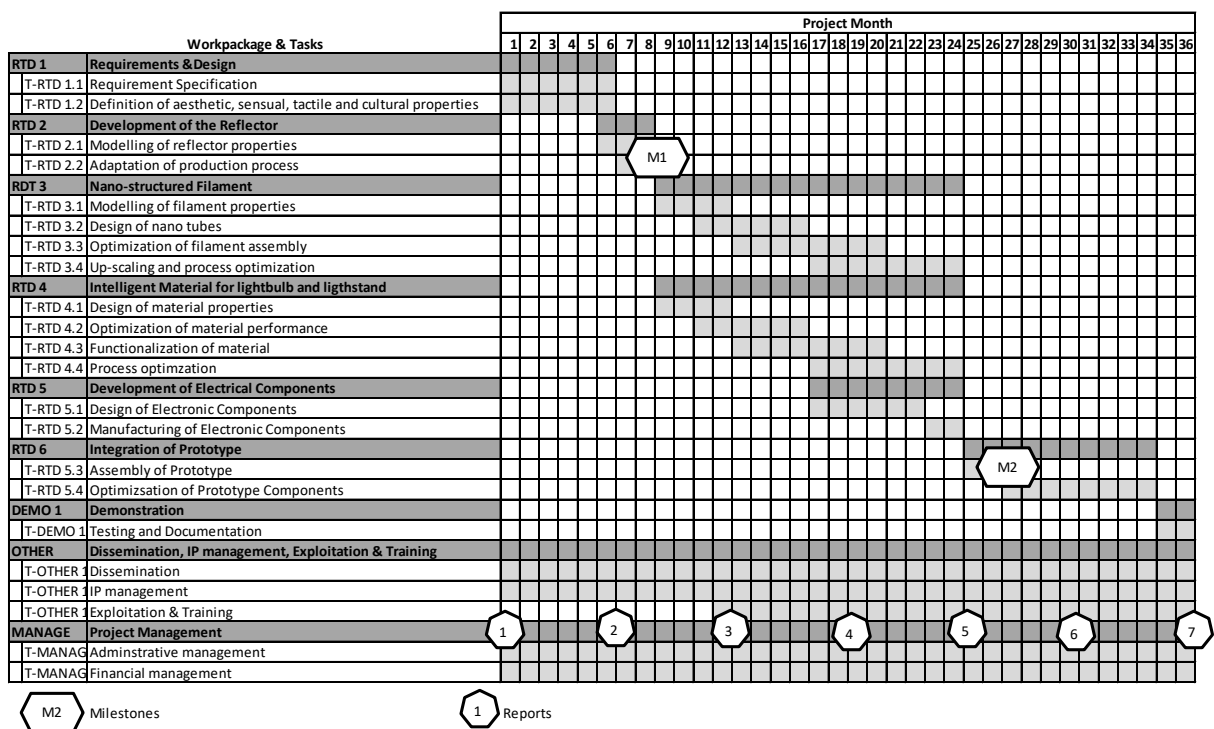


Figure 25 Gantt chart for the Aladdin's lamp project

## 6.4 Work package description

The work package description explains in detail how you will produce the result of the work package, who is involved, who is responsible, how much effort it takes and how you document the result. Check on the template for the work package description in the official documents. Should it have a specific format? How long may it be? Are there page limits to obey?

If no structure is provided you can structure it as shown in Figure 26:

- Situation at the starting point: Which results, knowledge, equipment and other are required at the beginning of the work package? Where do they come from, e.g. generated in a previous work package, literature, sub-contracting, already existing knowledge...?

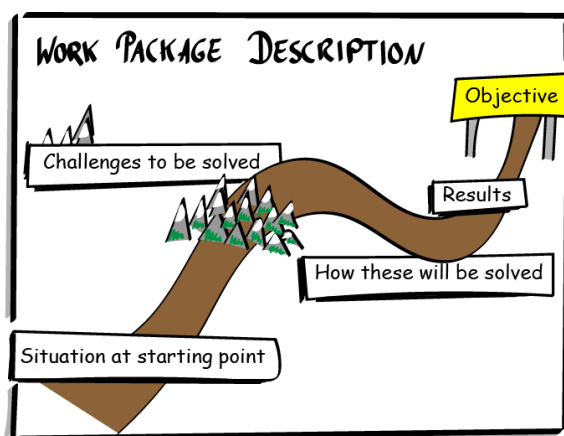


Figure 26 Work package description

- Challenges: What needs to be done to achieve these objectives? Which challenges do you expect?
- Solutions and alternatives: Which approach will you take? Which assumptions do you make to solve the challenges?
- Results: Which results will you have at the end of a task? What does it look like? What does it need to fulfil to contribute to the success of

the project?

If you have developed your work plan with the Disney method, you can now pick up your detailed cards. If you do not have any specific format, you can adapt the Horizon 2020 work package description template to your specific needs (see my example in Table 2 based on the format for research and innovation actions in Horizon 2020). For structuring the text, I propose the following:

- Objectives of the work package;
- Task 1;
- Situation at the starting point;
- Challenges to be solved;
- Steps to be done to solve them and
- Concrete results obtained at the end of each task.
- Task 2: proceed as above

The example text would still need to be improved by making it more specific and measurable.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                       |                                      |    |    |    |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------|----|----|----|------|
| <b>Work package number</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | RTD 1                 | <b>Start date or starting event:</b> |    |    |    | PM 1 |
| <b>Work package title</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Requirements & Design |                                      |    |    |    |      |
| <b>Participant number</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | P1                    | P2                                   | P3 | P4 | P5 | P6   |
| <b>Short name of participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |                                      |    |    |    |      |
| <b>Person/months per participant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |                                      |    |    |    |      |
| <b>Objectives:</b><br><p>In this work package, all partners are involved. As it is the starting work package, the partners need to get to know each other and the different viewpoints from scientist, technicians and designers. Bringing together all perspectives, the overall objective of this work package is to specify the technical requirements and the design parameters for Aladdin's lamp, so that</p> <ul style="list-style-type: none"> <li>• the interests of all project partners are considered by the development of Aladdin's lamp in order to ensure scientific and industrial relevance of the project results;</li> <li>• the fundamental working questions are clarified;</li> <li>• parameters against which scientific and technical achievements can be evaluated are specified;</li> <li>• the partners mutually get to know the way of thinking, the needs and interests as a basis for fruitful collaboration between scientists and designers as well as knowledge transfer.</li> </ul> |                       |                                      |    |    |    |      |
| <b>Description of work (where appropriate broken down into tasks), lead partner and role of participants</b><br><p>This work package is led by P2.</p> <p><b><i>T-RTD 1.1 Requirement Specification (Task leader P2, all partners involved)</i></b></p> <p>P2 will moderate a workshop at the kick-off meeting to collect the parameters, which are of interest for the individual partners to consider for Aladdin's lamp. The results of the workshop will be gathered in an</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                       |                                      |    |    |    |      |

Excel-table, which will be sent to all partners. Based on this each partner will internally discuss the individual objectives within the project and define their specific interests. According to these, each partner will formulate his requirements. In a second workshop led by P4 and P2 all partners will convene to discuss objectives, interests and requirements. In interactive sessions, they change perspective to experience the world of the scientist, technicians and designers. The results will be gathered and fused to a global picture of Aladdin's lamp. Each partner will have the time to evaluate the results. In a third workshop, the requirements to Aladdin's lamp will be agreed on.

***T-RTD 1.2 Definition of aesthetic, sensual, tactile and cultural properties (Task leader P4, all partners involved)***

Based on the specified requirements the design parameters will now be defined. In order to do so partners P4 and P5 will organise a fourth workshop for all partners looking into the creative world of design. This workshop targets at specifying the aesthetic, sensual, tactile and cultural properties of Aladdin's lamp. The objective is that scientists and technicians understand the vision of the designers. Following this workshop, technical and design parameters and limits will be derived as demanded for by the properties. The partners will balance both technical and design limits in order to specify the development parameters.

**Deliverables (brief description and month of delivery)**

|           |                      |                                                                                                                       |      |
|-----------|----------------------|-----------------------------------------------------------------------------------------------------------------------|------|
| D-RTD 1.1 | Specification Report | This report contains the final technical and design specifications used as target parameters for the development work | PM 6 |
|-----------|----------------------|-----------------------------------------------------------------------------------------------------------------------|------|

**Table 1 Example for a work package description**

## 6.5 Risks and contingency plans

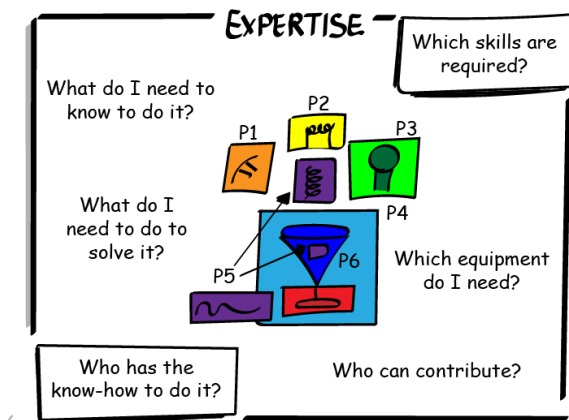
Each research project has risks. To be successful it is important that you know the risks and mitigate them already before they occur. If you consider this in your proposal it shows that you have a clear understanding of what you are doing. You can do this as follows:

- What may happen (in your project, not related to your project but inside your organisation or partner organisation, external e.g. on political or societal level) which may threaten your work?
- How are you going to deal with these risks? What can you do, if they come to pass?
- If you anticipate technical or scientific risks: Which alternatives are there, which you could follow within in project instead of your proposed way?
- If there are risks which threaten your project from outside: What exactly needs to happen, that you still can implement your project?
- Which project management activities are targeted to mitigate your risks? Think about milestones, working processes, decision-making or advisory groups or boards.

## 6.6 Expertise and complementarity of the consortium

Even if the project is designed for an individual fellowship for one researcher only, the identification of the necessary expertise and resources is an important step towards winning a grant:

- Being able to draw on the necessary expertise for completing the project successfully is a key for obtaining a grant as it makes the project realistic.
- The complementarity of the project partners with respect to expertise is a key to a winning project team. This requires that each partner is in the consortium for tasks only this partner can do. This reduces competition between the partners and thus it reduces conflicts.
- When applying for an individual fellowship or a grant to fund your own research group, your expertise is the key to the success of the project. Evidence for this expertise is provided by the way how you write about the state-of-the-art and the work plan, your curriculum vitae, the preliminary work and the list of publications.



**Figure 27 Identifying necessary expertise**

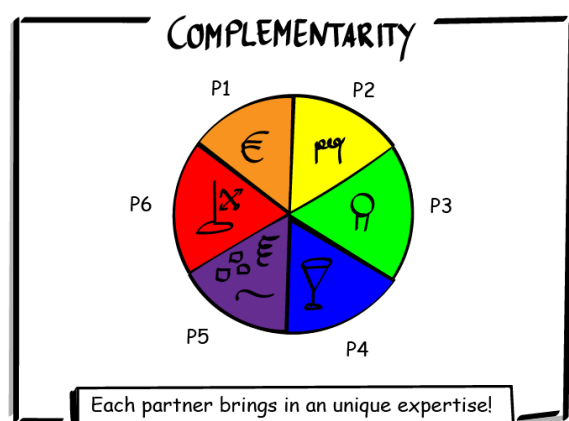
In order to identify the expertise necessary for successfully completing the project, you should ask yourself the following questions (Figure 27):

- What do I need to know to do it?
- Which equipment do I need?
- What do I need to do to solve it?
- Who has the know-how to do it?
- Who can contribute?

Some of these questions are already known from previous steps.

If you are the only applicant, you can now identify a good structure for your self-portrayal and the list of publications to present your expertise in the best light.

If you want to collaborate you can now identify and approach your project partners with a summary of the project idea. This comprises a first draft of objectives and work plan to explain the role of the new partners. If they agree, you will be able to integrate them in designing the details.



**Figure 28 Complementarity of a consortium**

It is also common practice that the project idea is generated around an existing consortium. This is due to the fact, that working together with partners you already know and trust reduces risks.

As said above one aspect of a winning consortium is the complementarity with respect to roles and expertise (Figure 28). This means that each partner brings in a unique expertise and has a unique role. Thus, tasks and responsibilities can be clearly defined, which will support the process of growing the consortium into an efficiently collaborating project

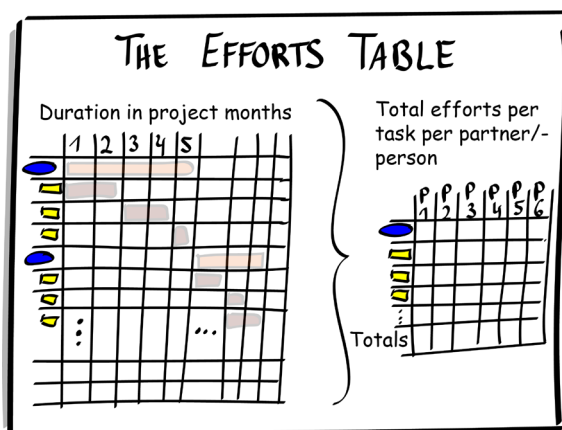
team.

Therefore, when writing up you clearly point out the complementarity and even visualise it. There are two different examples in this document:

- In Figure 27 the complementarity of the consortium is visualised in complementary colours. Their contribution or role of the individual partners is coded in the sketchy visualisations used throughout the proposal for denoting the different parts of the research project. In this figure, the focus lies on the aspect of making the project round, of running it successfully with the consortium. The use of colours however is tricky, as you should expect the proposal to be printed in greyscale.
- In Figure 28 the complementarity of this consortium is related to the overall project vision.

It is a question of context and of visual language, what is the best way in “marketing” the expertise of the consortium.

## 6.7 Planning the efforts



**Figure 29 The efforts table**

As soon as we know the detailed activities and who will be involved in the project, we can start the efforts planning. This is best done in an efforts table (Figure 29). Each work package presents a row in the table, below which the tasks are sorted according to their numbering. For a good handling of the table the full titles of work packages and tasks should be presented.

For each partner or person in the project, a column is reserved. Now you can enter the efforts either in person hours or person months for each individual task. The totals of the tasks belonging to a work package are then entered in the work package line.

This way we can use the table for

- presenting an overview on the work packages by hiding the task rows or
- showing the detailed efforts by showing everything.

Besides planning the efforts of each partner in the individual tasks, this table is a nice tool for visualising the role and the involvement of the project partners. The cell of the partner responsible for this task

| Workpackage & Tasks |                                                                   | Efforts in pm |    |    |    |    |    | Total     |
|---------------------|-------------------------------------------------------------------|---------------|----|----|----|----|----|-----------|
|                     |                                                                   | P1            | P2 | P3 | P4 | P5 | P6 |           |
| <b>RTD 1</b>        | <b>Requirements &amp; Design</b>                                  | 2             | 6  | 2  | 6  | 4  | 2  | <b>22</b> |
| T-RTD 1.1           | Requirement Specification                                         | 1             | 4  | 1  | 2  | 1  | 1  | 10        |
| T-RTD 1.2           | Definition of aesthetic, sensual, tactile and cultural properties | 1             | 2  | 1  | 4  | 3  | 1  | 12        |
| <b>RTD 2</b>        | <b>Development of the Reflector</b>                               | 5             | 1  | 1  | 2  | 1  | 1  | <b>11</b> |
| T-RTD 2.1           | Modelling of reflector properties                                 | 2             | 1  | 1  | 1  |    |    | 5         |
| T-RTD 2.2           | Adaptation of production process                                  | 3             |    |    | 1  | 1  | 1  | 6         |

**Table 2 Example efforts table**



should be marked differently, e.g. with bold font or shading.

Table 2 shows the planning of efforts for the Aladdin's project with respect to the involvement of each partner. Highlighted in grey is the partner being responsible for the respective task.

If we compare this table with Figure 25 we can see that RTD 1 is the only work package running for the first 6 months. From this point of view, it is positive that all partners contribute to both tasks. This helps the partners to get to know each other and to get involved in the project. To understand this, just envision the timeline from the first idea to the start of the project. Those who wrote the proposal are usually the only ones that have a clear idea of the overall project idea and setup. Though some partners will play a leading role only later in the project, it is important to involve them right at the beginning.

An aspect, that plays an important role on the level of European funding, is the networking between different organisations from different countries. You might consider this as a contribution towards building Europe. Building up relationships includes working together which in turn requires interaction. You will support this by having different partners working together in a work package. A reviewer can see this from the distribution of efforts by partners.

Moreover, one aspect of being motivated to contribute to the project is the question of responsibilities. Someone who is responsible for a specific work usually will show a higher commitment. Therefore, each partner should take over some responsibility. This is easy to assess for a reviewer if you indicate it in the efforts table.

## 6.8 Costs for consumables, equipment and travel costs

Costs for consumables and equipment are usually funded. You should list them according to the rules and format as specified by the funding programme. You need to justify the costs. This can be done by

- Listing individual items and their costs,
- By listing the meetings and journeys for the travel budget
- By providing quotes for the equipment and services
- By using lump sums whenever provided.

Equipment and services or sub-contracts need to be specified in detail and are usually justified with quotes. You will also have to discuss with your administration, whether you are allowed to put them and what to observe.

## 6.9 From efforts and equipment to calculating the budget

With a complete effort table and our list of equipment and consumables, we have completed the most important steps for calculating the budget of our project. The following instruction is rather generic and needs to fit to the requirements and financial rules of the funding programmes. Usually templates are available, showing in which way the budget should be calculated and presented.

Before you start, you should follow the checklist to determine how it works:

- Which costs are eligible?



- How are personnel costs/monthly rates calculated? If they are based on lump sums, are there any predefined rates?
- What is the minimum and maximum budget?
- What is the minimum and maximum funding limit? What is the funding rate?
- Which cost categories exist? What are the respective funding rates?
- Is there a template? Is it complete?

Be aware, that project costs and the budget, which you are going to submit with your grant proposal are two different things: The budget is usually smaller than the real project costs, as some costs are not eligible. Therefore, you should clarify, that your host organisation is going to cover for all the costs, which are not going to be funded. This also holds true for funding gaps arising from funding rates less than 100 per cent.

## 7 Writing a proposal that can be understood easily

Our first idea is that formulating the proposal is a challenge:

- There are page limits or limits on characters for the individual chapters.
- The grade of detailing is not clearly defined.
- The partners have to contribute in time.
- Contributions of others have to be inserted.

This perception is our own perception and in mastering this challenge, we often forget about the following:

- The proposal will contain our research objectives, the fundamental questions, the motivation, the plan and much more. If we would read it aloud, we would easily understand what it is. It is a monologue on what is important for us and what we would like to have funded.
- Our monologue has an audience. It is for one or two reviewers who will have some time for reading the parts, which are of interest for them. Unfortunately, they do not hear our voice, nor do they have the possibility to clarify, whether they understood correctly.
- Therefore, how they understand what we have written down is in their sole responsibility. What they will understand depends very much on how they see the topic of research, the world, what they have eaten for lunch...

Once our proposal has been submitted, we have no means of influencing what happens in the heads of the reviewers. Therefore, we must take care of it when writing the proposal.

Before we do so, let us see, what happens in the communication process between the proposal writer and the reviewer:

- The proposal writer has a picture of his project in his mind.
- Before this picture is transformed into a full text, some filters will be active, sorting out what is important enough to be formulated in words.
- The reviewers read the proposal. Again, filters process the information, and only the information which is important to the reviewers will survive. This will be compared with their idea of what the funding authority wants to fund and what makes a potentially successful project.
- If the information fits in well, the proposal will be evaluated positively, if not the drawbacks will dominate the picture. Both will influence the reviews.
- The proposal writer will receive the evaluation result with more or less details. The drawback outlined in the review reports are compared with the own perception of the research project proposal. Very often this ends in being upset, because for the proposal writer it is there and clear (but it was not formulated explicitly in the text).

We can see from this consideration, that misunderstanding is a natural feature of a reviewing process. Therefore, the true challenge in communicating our project efficiently lies in communicating our image of our proposal in such a way, that what the reviewer understands fits nicely to his own subjective image

of what the funding authority wants. In other words, we write for the reviewer and not for the funding agency.

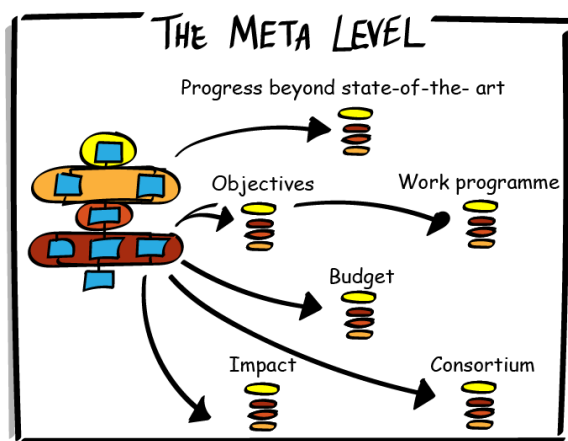
With respect to understandability we find some guidance in Langer, et al. (2019)<sup>4</sup>. They have shown, that understandability is determined by simplicity, structure, shortness and stimulation:

- Simplicity refers to a simple presentation of the content in terms of language and grammar, defined terminology, conciseness and clarity.
- Structure refers to a clear line of argument, clear arrangement, differentiation between important and unimportant aspects in text and visual structure of the text.
- Shortness refers to being concise but still elaborate enough to enable the reader to follow the line of argument easily.
- Stimulation refers to those aspects, that make the text interesting to read, like examples, visualisations.

But how to get there? The following methods will help to gather all the details, before writing the full text.

## 7.1 Visualize your project

Visualisations are key to explaining something complex in an easy way. They should be self-explanatory and simple, so that they can be understood much faster than reading the text. The idea behind my work with visualisations is to make them as concise as possible for providing a frame for structuring the text. As a lot of decision-making is involved in the elaboration of a visualization of the overall objective, the research design, the progress beyond the state-of-the-art, the approach, the complementarity of the consortium, I like to work on visualisations long before I start writing the full text.



**Figure 30 Using the meta level for structuring the text**

You can use a good visualisation as a “meta level” of your project. It is then something like the skeleton of a human: You can easily recognise the different parts of the body, but you do not have all the details around the bones.

Very often parts of the research design or PERT-diagram with concise descriptive titles can be used as the meta level.

Once you have visualised the meta level of your project, you can use it to structure your text (Figure 30). In our example of Aladdin’s lamp, we have already encountered this to visualise the contributions of the project partners. Now, we use the same strategy to

<sup>4</sup> Langer, Inghard, Schulz von Thun, Friedemann und Tausch, Reinhard. 2019. *Sich verständlich ausdrücken*. 11. Auflage. München : Ernst Reinhardt, GmbH & Co KG, 2019.

structure the line of argument in any part of the proposal. For example:

The overall objective of the proposal is Aladdin's lamp meeting some specific parameters. This will be achieved by

1. A mirroring helmet which fully-reflects and focuses the emitted light;
2. The assembly of nano-tube LEDs on a filament;
3. A liquid crystal glass for the light bulb which partially turns opaque for light effects;
4. A solar stand which can be twisted and bended and which changes its haptic and optical characteristics with respect to customized functions reflecting fundamental physical laws;
5. An innovative electrical concept maximising the light performance of the lamp while minimising the consumption of electricity.

Following this line of argument, we now present the contents of the other chapters in the same structure. The chapter on progress beyond state-of-the-art is structured according to

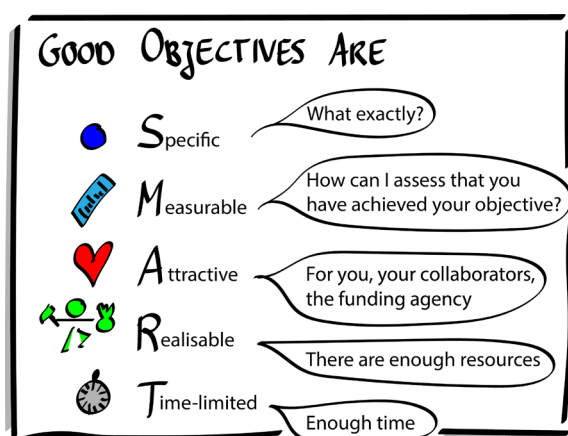
1. Mirroring Helmet
2. Nano-structured Filaments
3. ...

The same is done with the other chapters.

## 7.2 Question your objectives

An objective (Figure 18) is a result, that will be achieved until a specific point of time in the project by eating up specified resources. If you connect this objective with a decision, which is done based on a checklist, you will have milestone. In a research project these decisions are often related to the questions "How exactly are we going to proceed with the research?". The information on results, time and resources is distributed over various sections of the proposal: the objectives, the work programme, the timeline and the budget.

It is old management knowledge that good objectives are SMART (Figure 31). There are different interpretations of the letters, we will use the following:



**Figure 31 Definition of SMART objectives**

- S for specific: What exactly is the result of the project? How can it be described in detail? Which features does it have?
- M for measurable: Which requirements will the result satisfy? How can someone else assess, that the result is there? Which parameters will describe the result?
- A for attractive: What exactly makes the result interesting for the researcher, the team, the organisation, the funding agency, the sponsor? What is the impact for the research community?

and for stakeholders? Which impact will result from the project?

- R for realistic: Is the project feasible? Are there enough resources in terms of knowledge, access to facilities, staff, budget for achieving the result in the given timeframe? Can all the anticipated problems be solved?
- T for time-limited: When will the result be delivered?

As the A is related to the impact, the R and T a result of the detailed planning, the S and M can be derived from asking open questions. Open Questions are questions that elicit more information whereas closed questions ask for a yes or no. They are very helpful for decision-making.

The easiest way to this:

- Write a first version of your objective.
- Now look at each word and ask open questions like “What does this mean exactly?”. Find the answer and ask more open questions for each word in the answer.

Asking questions is easy and answering the questions will be difficult. Jot down your answers in keywords. They will contain information which you will write down in other sections than the objective section.

### 7.3 The innovation or progress beyond the state-of-the-art

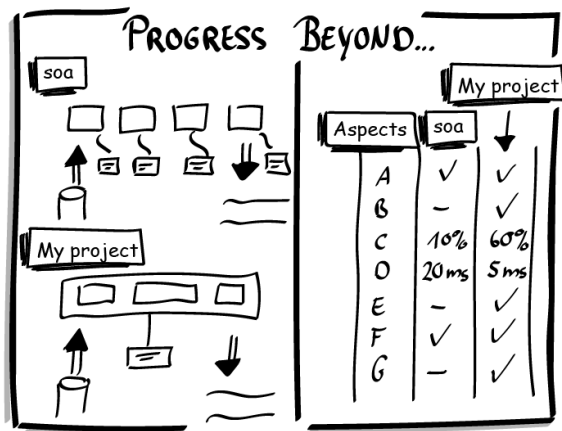


Figure 32 Progress beyond the state-of-the-art

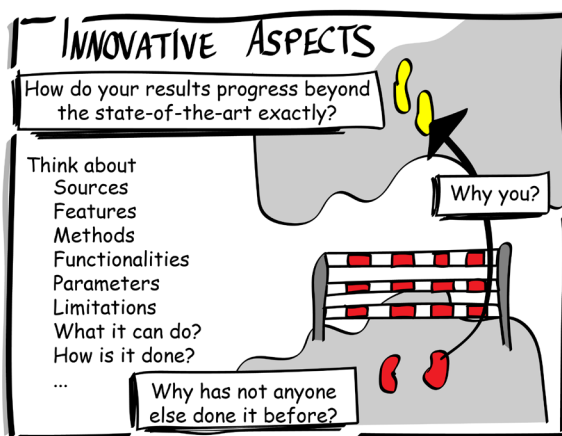


Figure 33 Innovative aspects

Probably the biggest challenge in matching the reviewer's view of the world is the description of the progress beyond state-of-the-art. However, this is at the very heart of any good proposal because it is the answer, why the fundamental question you attempt to answer is attractive to the funding authority, the community, yourself, the industry, the public, the society etc.

It is essential to be very concise and clear and still provide enough overview for showing your scholarship.

This can be done by:

- Naming the meaning of the state-of-the-art in the context of the proposal by providing numbers, figures etc. against which the performance of the respective project result can be measured.
- Explaining how the progress will be achieved and what needs to be done to get there.
- Highlighting the risks on the way in the respective chapter.
- Providing two images comparing the state-of-the-art and the project result.

The last and the first option are shown in Figure 32.

For developing a good understanding of the actual innovation, the barriers and why you are the right person, you can use the questions of Figure 33:

- How do your results progress beyond the state-of-the-art exactly? How does you approach progress beyond the state-of-the-art?
- Why has not anyone else done it before? Which barriers are there exactly?
- Why are you the right person/consortium to achieve this?

The answers to these questions will then inspire the different sections of the proposal.

## 7.4 The impact

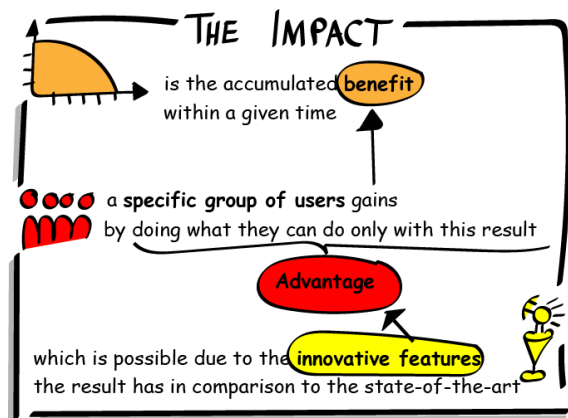


Figure 34 Definition of the impact or outcome

Something difficult to handle is the impact of the proposal. In many proposals for European funded research projects there is a distinct chapter reserved for the description of the impact. But what exactly is the impact?

I define it as shown in Figure 34:

The impact is the quantified benefit a specific group of users achieves within a given time by exploiting the advantages of the result enabled by specific features the result has in comparison to the state-of-the-art.

Providing a convincing impact is difficult. You can derive it step by step by writing down (Figure 35:

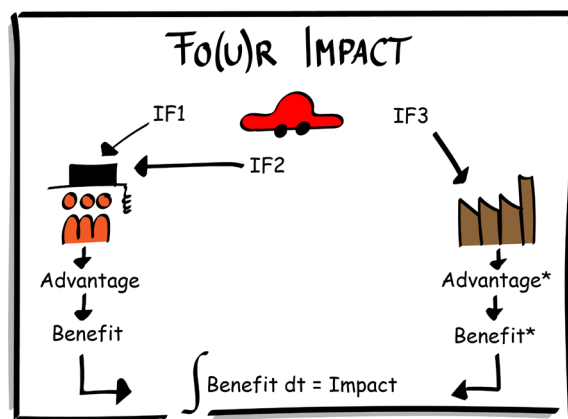


Figure 35 Method for identifying the impact

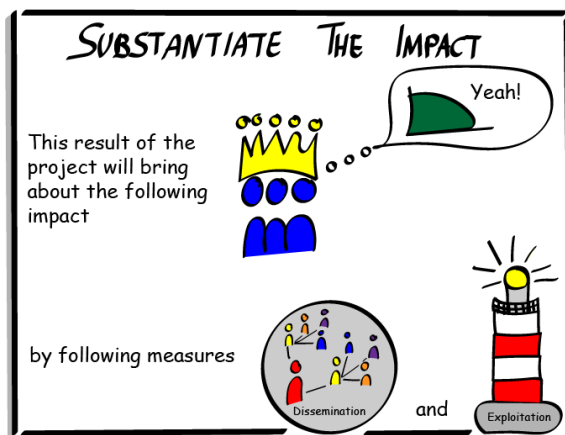
exploitation for a defined period of time.

- Specific features which differ from state-of-the-art;
- Your target groups;
- The advantages each target has, which is created by all or a subset of these specific features;
- The (quantified) benefit, which the users will have if they exploit the advantage;
- The impact you will achieve if you look at this

For the difference between a feature, advantage and benefit, look at the definition of Tim Taxis<sup>5</sup>:

- A feature is a property, functionality or specific part of your result. This could be material properties, features of a model, the application area of a function, the sequence of steps in a process etc.
- An advantage names how the features of your result are of help for the specific user. Your features of a model e.g. help a meteorologist modelling the atmosphere to approach more realistic parameter regimes.
- The benefit is a concrete answer to the needs of the specific user. It is the translation of the advantage into the benefit. Thus, in the case of our example above, the features of the model

<sup>5</sup> Taxis, Tim. 2011. *Heiß auf Kaltakquise - So vervielfachen Sie Ihre Erfolgsquoten am Telefon*. Freiburg: Haufe-Lexware GmbH & Co. KG, 2011, copyright protected



**Figure 36 Substantiate the impact**

attractive, because it answers to the specific needs of the funding authority. These needs are defined by its vision and mission. This could be to push basic research, the competitiveness of the European Union, to push applied research in a specific field of interest etc. Have a look at your mind map on the framework: There it should say what it is exactly.

One key to success in writing a successful proposal is a precise language. This especially holds true when substantiating the benefit and impact of the project. If we are clear about this, we can answer to the impact listed in the call documents (Figure 36):

- For a specific group of users this project will bring about impact XY because with feature 1 this result will bring about the advantage 1. Therefore, by doing step 1, step 2 and step N this group of users will have the following quantified benefit.
- To make this argumentation plausible it is helpful to add a supporting calculation.

Impact will only be convincing if you integrate dissemination and exploitation activities which will support your target groups in picking up your results.

## 7.5 Political and societal impact

It probably feels quite strange to be asked to formulate the impact the project has on e.g. the economic strength of the European Union, science in general, the environment, society or whatever is asked. It is a systemic approach to an institution investing money into a research project. The answer to this question is one of the aspects that make your project attractive for funding. Therefore, it is very important to formulate the political and societal impact in a convincing way.

To derive it you can consider the chain of cause and effect:

- If I have A. What can I use it for?
- If I have achieved B with A, who will use it?
- If x people use B, what effect results?
- What further consequences has this effect?
- What is this useful for?

will create only a benefit, if the meteorologist interested in global models really requires a more realistic parameter regime. If the physical phenomena which can be incorporated in the model have a negligible effect on the overall state of the atmosphere, there is no benefit. If it does have a substantial effect, the meteorologist will certainly have a benefit.

Differing between features, advantages and benefit helps in the selling process to make the offered product relevant for the customer. In our case of proposal writing we use it to make the project result



Usually a qualitative discussion will be enough. Anyway, if you can do a supporting calculation which is somewhat realistic, it will be a good idea to add it. Before you do so, ask yourself if it will induce conflicting emotions or scepticism on behalf of the reviewer. If yes, it is probably better to stay with your qualitative discussion.

## 7.6 Structuring proposals

Now you have all the details for writing your proposal. You will obtain the correct proposal structure by following Figure 37.

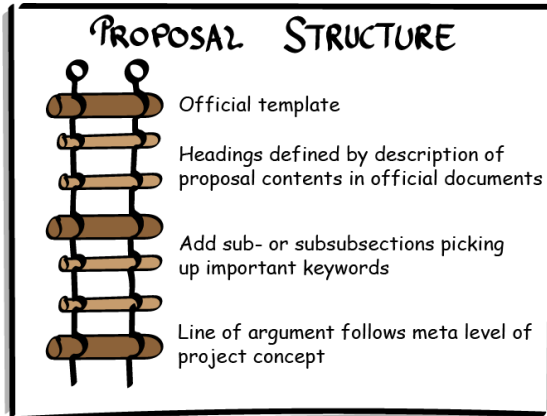


Figure 37 Proposal structure

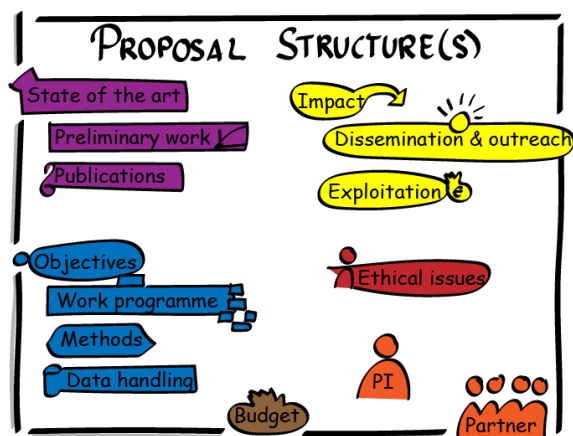


Figure 38 Potential chapters in proposal

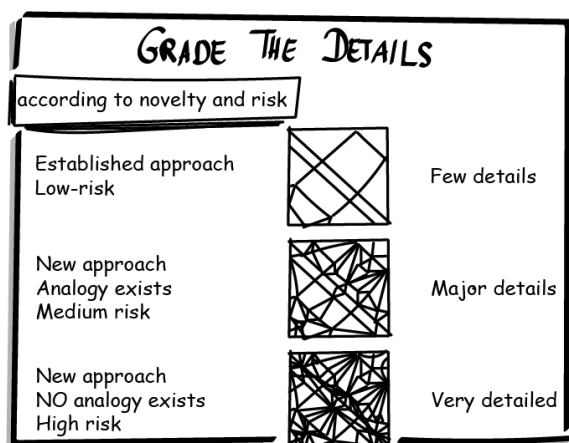
deeper into your text as needed. Therefore, it is helpful to pick up important keywords for further structuring your proposal and to follow the same structure of argumentation in the different parts of the proposal. For this you can use the meta-level of your project or the essential parts of your research design.

Figure 38 shows the standard chapters or sections which you can find in many proposal templates. The purple, blue, red, orange and brown parts represent the proposal template for the DFG proposals without the modules. The yellow part is a chapter that can be often found in funding programmes targeting at applied research. Proposal templates or supporting documents will give you some ideas which information should be given in which section.

## 7.7 Grade the details

When working out the structure of your proposal and creating your proposal template, one important parameter is to find out which limits on pages and characters have been defined. With the restricted page limits, it lies at hand to vary the detailing to cope with the limited space. This is not only a means of

1. First check among your official documents for the official template for the proposal.
2. If there is no official template, the structure of the proposal including the headings will be defined somewhere in the official documents. For European funded research projects in Horizon 2020 the Guide for Applicants provides further information. It is published for each call as an individual document and contains a detailed explanation of the proposal submission process, the evaluation process and in the annexes a detailed description of the proposal. If there is no template, copy the structure into your own template including the text below the headings.
3. Use the description of what is expected under each heading to further structure your text with subsections or sub-subsections. Now you have the basic template with which you can start to work.
4. A clear structure supports you in writing a good proposal and it helps the reviewer to find his way around your proposal. Some reviewers will look at things important for them first and will then dive



**Figure 39 Grade the details**

getting all necessary contents inside the proposal, it is also the art of conciseness to save the reviewer from dispensable details.

Figure 39 presents my thumb rule, which is also applicable to any scientific publication. Details should be graded according to the novelty of the presented concept and the risk connected to the respective steps of development:

- If the concept is established and the development is low-risk, few details are sufficient. The objective of these details is to show the expertise in this field. The naming of

the concept, its meaning for the project and its limitations with respect to the objective should be presented with a citation of the publication recognised as the fundamental publication with respect to this topic.

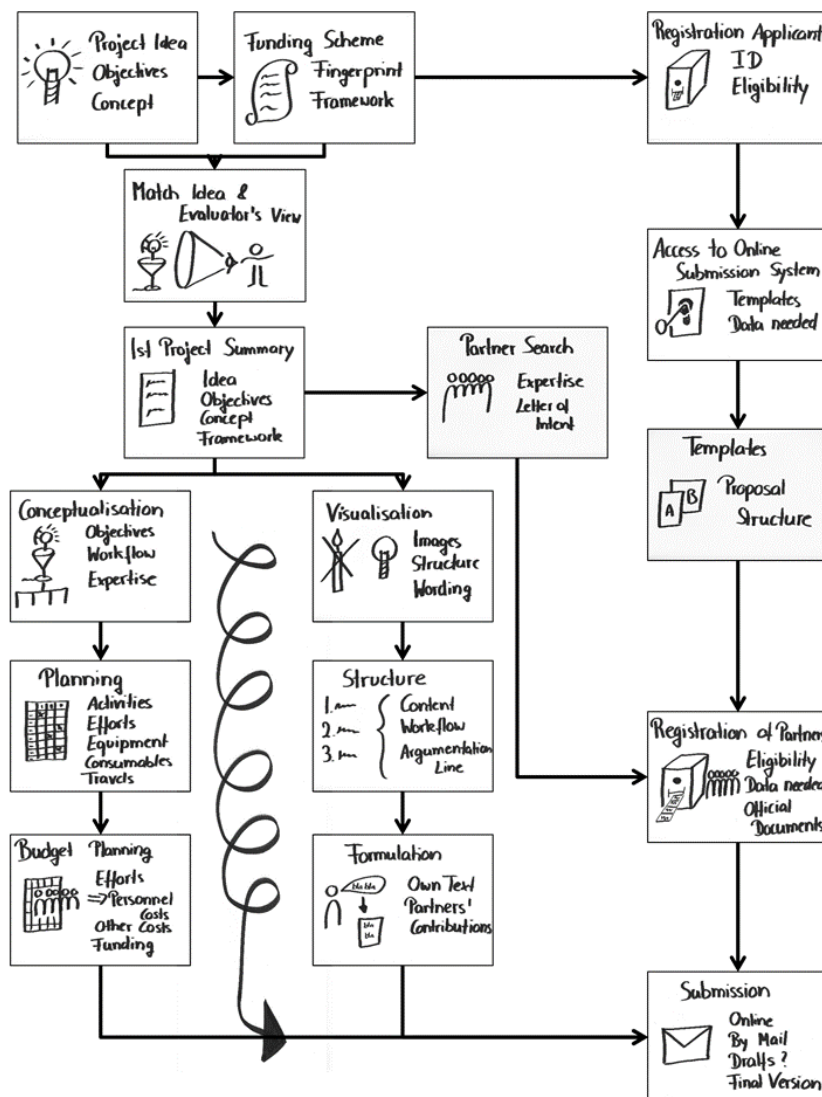
- If the concept is new in your field of research but it is an established concept in another field of research, you should provide the major details. It is good to have a feeling on the risks connected to the adaptation in your research projects. Are there analogies and solutions in the other field of research? If yes, name them and provide some citations for reference.
- The presentation of a completely new concept, which has no analogy in another field of research or development steps which are very risky should be discussed in detail. Naming risks and contingency plans shows that you have a deep insight into your area of research and that you have a deep understanding of what you are attempting. Therefore, look at the crucial details and describe what you intend to do to care about the connected risks. Provide references to the publications of well-known experts to support your argumentation.

## 8 Coordination of the proposal writing and submission process

The previous chapter focused on the details of designing your proposal and formulating the text. This chapter is dedicated to the coordination of the proposal writing and submission process. Coordination means to arrange for the different activities to be done in such a sequence that the overall result will be achieved. In this case it means to conclude all the activities in such a way, that the full proposal with all necessary administrative data can be submitted before the deadline.

As the elaboration of a good research proposal requires some forth and back, iterations here and there, many things need to be done in parallel. I have tried to sketch this for the submission of a collaborative proposal to the European Commission in Figure 40. The left includes the actual project design and proposal writing process. The right shows what needs to be done on top. Depending on the structure of the consortium you might want to replace partner search with search for a hosting institution and add a

work package for gathering supporting documents.



If you imagine yourself a timeline from top to bottom of Figure 40 you see that I start the registration more or less already at the beginning of the process. It is good to know oneself around the electronic submission platforms and to conclude the administrative part as soon as possible. You will get important information on templates, contents and data needed by looking closely at the system. Expect, that it will take some time to gather the data especially if administrative departments of the partner institutions are involved. And mark: There are still some funding programmes around which require an official signature of the heads of the involved organisations in order to activate the electronic submission.

**Figure 40** From idea to submission of a collaborative European funded research project

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